# Input Form (IF)

## Score: 3/5 — Moderate gaps | Confidence: high

***Benchmark:*** *MMLU* | ***Context:*** *Non-Arab Tourists and Expats in Eight Arab Countries*

**Input Form definition:** Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

MMLU's text-only English MCQ format [Q3, Q56, Q71] matches the deployment's primary interaction modality (English text), which the elicitation rates MODERATE priority. However, the Arabic-learning secondary cohort cannot be evaluated by MMLU at all, and MENAValues research [WEB-11] documents cross-lingual value shifts indicating that English MMLU performance does not predict Arabic-prompt behavior. Format brittleness (e.g., separator-token sensitivity [Q105]) adds modest external-validity concern but is not region-specific.

| ID | Evidence |
|---|---|
| Q3 | “We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings.” (p.1) |
| Q56 | “We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject.” (p.6) |
| Q71 | “Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format.” (p.7) |
| Q105 | “If we remove the </s> from the input, accuracy declines by several percentage points.” (p.12) |
| WEB-11 | MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154) |

## Strengths

- English text-only MCQ format aligns with the primary user cohort's interaction modality (Strength 2; D6, D7)
- Zero-shot/few-shot evaluation paradigm [Q3] matches the open-domain generalization required for tourist/expat queries (Strength 4)
- Formal English register is appropriate for educated adult tourist/expat users [Q56]

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| D6 | A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario |
| D7 | A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure |

## Checklist

### Checklist item definitions:

| ID | Assessment Question |
| --- | --- |
| IF-1 | Compare signal distributions between source and target |
| IF-2 | Check if regional infrastructure supports data capture specs |
| IF-3 | Identify domain-specific form differences |
| IF-4 | Document form mismatches harming external validity |

### Checklist responses:

**IF-1**: MMLU is exclusively English text [Q71]; primary deployment interaction is English text — match. Arabic-script and dialect inputs from the secondary cohort are unsupported by MMLU and require separate benchmarking [WEB-1, WEB-2].

**IF-2**: Regional infrastructure not a concern — tourists/expats use standard text-based devices; deployment YAML notes this is not a connectivity gap context.

**IF-3**: DialectalArabicMMLU [WEB-1] covers Moroccan, Egyptian, Emirati, Saudi dialects (4 of 8 countries) but not Kuwaiti, Jordanian, Palestinian, or Lebanese dialects; format-level dialect sensitivity is unaddressed in MMLU.

**IF-4**: Form mismatch is moderate: English MCQ matches primary cohort; Arabic-language input form is entirely unevaluated by MMLU. Cross-lingual value shifts [WEB-11] mean MMLU scores cannot be assumed to transfer to Arabic prompts.

| ID | Evidence |
| --- | --- |
| Q71 | "Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7) |
| WEB-1 | DialectalArabicMMLU covers five Arabic dialects (Syrian, Egyptian, Emirati, Saudi, Moroccan) with substantial cross-dialect performance variation (https://arxiv.org/abs/2510.27543) |
| WEB-2 | ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/) |
| WEB-11 | MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154) |

## Information Gaps

- Tourist/expat device interface preferences (mobile vs. desktop) not searchable at required precision (FV10)
- Whether Arabic-learning users predominantly prompt in MSA vs. dialect

## Requires Expert Verification

- Empirical sizing of the Arabic-prompting sub-cohort relative to the English primary cohort

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## Remediation

**Gap 1: English-only format does not evaluate the Arabic-learning sub-cohort.**

**Recommendation:** Add DialectalArabicMMLU [WEB-1] for Moroccan/Egyptian/Emirati/Saudi dialect coverage and ArabicMMLU [WEB-2] for MSA. Note that Kuwaiti, Jordanian, Palestinian, and Lebanese dialects remain unbenchmarked — flag as residual gap.

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